AHARADA EDUCATION

Excellence in Aviation, Aerospace & Advanced Computing Technologies

OFFICIAL CURRICULUM STUDY MATERIAL & LECTURE NOTES

COURSE & SUBJECT SPECIFICATION

Course: B.Sc Aeronautical Science
Subject Code: MBA - AI105
Subject Name: Data Base Management & SQL
Semester: Semester 5
Target Unit: Unit 1
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Unit 1: Data Base Management & SQL

Subject: Data Base Management & SQL (MBA - AI105)

1. Unit Syllabus & Module Breakdown

Introduction to Data, Information, and Databases; File Processing System vs DBMS; Advantages of DBMS; Database Architecture; Components of DBMS; Three-Schema Architecture; Data Models; Relational Database Model; Database Users; Database Administrator (DBA); Introduction to MySQL, PostgreSQL, Oracle, SQL Server; Applications of DBMS in AI, Data Analytics, Finance, Healthcare, Retail, and Supply Chain.

2. Core Learning Objectives & Key Concepts

- **Topic 2.1:** Introduction to Data, Information, and Databases
- **Topic 2.2:** File Processing System vs DBMS
- **Topic 2.3:** Advantages of DBMS
- **Topic 2.4:** Database Architecture
- **Topic 2.5:** Components of DBMS
- **Topic 2.6:** Three-Schema Architecture
- **Topic 2.7:** Data Models
- **Topic 2.8:** Relational Database Model
- **Topic 2.9:** Database Users
- **Topic 2.10:** Database Administrator (DBA)
- **Topic 2.11:** Introduction to MySQL, PostgreSQL, Oracle, SQL Server
- **Topic 2.12:** Applications of DBMS in AI, Data Analytics, Finance, Healthcare, Retail, and Supply Chain.

3. Detailed Chapter Notes & Concepts

3.1 Introduction to Database Management Systems (DBMS)

A DBMS is system software for creating, managing, and querying databases. Compared to legacy file systems, DBMS provides structured data independence, reduces data redundancy, enforces concurrent access control, ensures ACID properties (Atomicity, Consistency, Isolation, Durability), and provides automated backup and recovery mechanisms.

3.2 Three-Schema Architecture (ANSI/SPARC)

The three-schema architecture partitions database systems into three abstraction layers:
1. External Level (View Schema): Individual end-user views of the data.
2. Conceptual Level (Logical Schema): Community view describing what data is stored and relationships among entities.
3. Internal Level (Physical Schema): Physical storage organization on disk, indices, and file structures.

3.3 Database Users & DBMS Applications

Key stakeholders include Database Administrators (DBAs responsible for schema definition, permissions, security, and

performance tuning), Application Programmers, and End Users. Modern relational engines include PostgreSQL, MySQL, Oracle, and Microsoft SQL Server, powering mission-critical workflows in Aviation, Airline Operations, Financial Analytics, Supply Chain, and Artificial Intelligence.

4. Review Questions & Self-Assessment

1. Explain the fundamental architectural distinctions between physical and logical data independence.
2. Discuss how the topics covered in this unit apply to industrial workflows in B.Sc Aeronautical Science.
3. Formulate standard case diagrams and explain key constraints discussed under Unit 1: Data Base Management & SQL.
4. Review previous university examination questions related to Data Base Management & SQL (MBA - AI105).